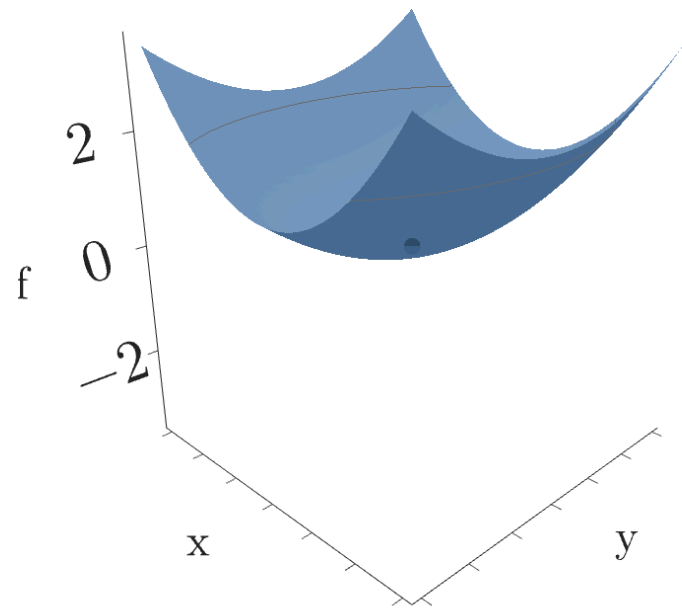
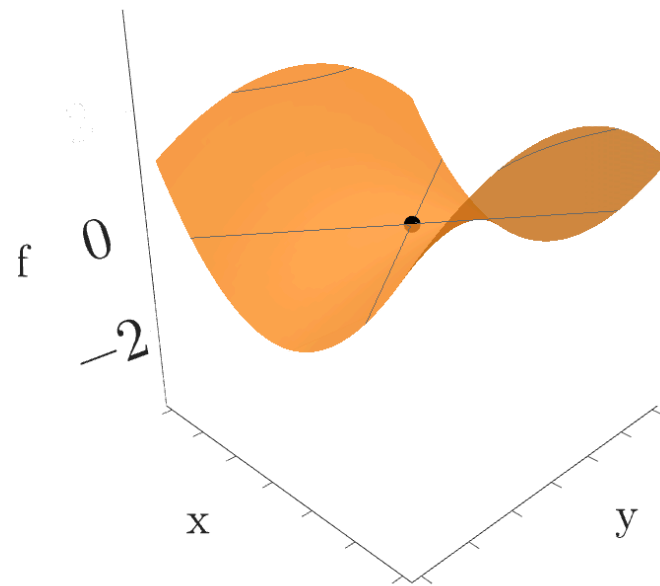


bowl:  $H = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$   
eigenvalues 2, 1



saddle:  $H = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}$   
eigenvalues 2, -1



cap:  $H = \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix}$   
eigenvalues -2, -1

